

PAPER_1763 — Hubble Bubble Underdensity = $\delta\rho/\rho = -F_{\text{TRZ}} \times \beta_i \times 5 = -30.1\%$ (resolves H_0 tension)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.48% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1457 (PAPER_132x-135x corpus) gives:

$$\delta\rho/\rho = -F_{\text{TRZ}} \times \beta_i \times 5 = -30.1\% \text{ (resolves } H_0 \text{ tension)}$$

Status: **0.48%**.

UQFF Closed Identity

$$\delta\rho/\rho = -F_{\text{TRZ}} \times \beta_i \times 5 = -30.1\% \text{ (resolves } H_0 \text{ tension)} \quad 0.48\%$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1457
- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)
- Calculator dispatch: `calculate_paradox({"paradox": "hubble_bubble_30_pct"})`

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